The B.S PRECISION CASTING

B.S Precisiom casting (BSPC) one of the group companies was established in 2025 at BHIWADI, INDIA, and is a well known firm in the NCR market in the areas of manufacturing of Precision Investment Castings that are raw as well as finished. The firm manufactures engineering components of the highest quality, by the process in which one disposable pattern produces one metal part.

The process which was initially confined to typical aerospace requirements is now used in all engineering fields to produce components of various degrees of integrity and complexity. The products are today available in volume and for the broadest spectrum of applications in condition with ready to use or very close to the final product for end use.

Today leading OEMs and other manufacturers have accepted this process as a solution of excess machining, welding, fabrication, assembling and other time and cost consuming processes.

BSPC willing to make and supply products with cost effectiveness, high productivity and with commitment of continual improvement is BSPC.

With a total installed capacity of Three hundred metric tons per annum, BSPC is proud of a well laid out modern plant.

Why BSPC?

- We are located at Bhiwadi – The biggest hub of investment castings of INDIA.
- Our system & manufacturing process are approved by leading notifying agencies that can increase confidence level towards the supplier selection and put us one step ahead from our competitors.
- We are out of few foundries having the approval and the ability for supplying product with precision machining.
- We know the requirements of domestic & international OEMs and your product will be also treated with same experience & expertise.
- We will support you to increase your customer satisfaction by providing value adding solution for your product for making it more cost effective with increasing productivity.

- We can also able to fulfil your requirements of raw or finished products made by forgings, sand castings & zinc pressure die castings from our group companies with single window operating, which can save your time, manpower, cost of logistic & more.

Promoter:

Mr. BHUPENDRA SINGH is the proprietor of this organisation. entrepreneur and by profession he is an Engineer. Sh. Bhupendra singh the main promoter of the Company has over Three years of experience in Steel Industry. He started his Career with Saraswati engineering works located in bhiwadi. Where he worked for three years as Steel Foundry engineer.. In 2025 Sh. Bhupendra singh formed ***B.S PRECISION CASTING*** and set up his own Steel Foundry having production capacity of 1000 M.T. per annum and since then the company is working successfully.

Name	Address	PAN No	Date of Birth	Net Worth (Rs. In Lacs)
BHUPENDRA SINGH	BDI sunshine city ,	PTHPS8760Q	18-08-2001	---
	Bhiwadi raj-301019			

The Group

Setting Industry best standards is the key vision of the BSPC Group as they manufacture a variety of products for an array of industries. With close collaboration with their valued clients, the BSPC Group develops and manufactures necessary commercial solutions in short time deadlines and deliver products as per specific client requirements.

Upgrading technology and global engineering focus areas help the Anjni Group understand their customer requirements. With an enviable global reputation, the Group has managed to consistently provide high quality standards with leading edge integrated solutions for their customers. With a dedicated and well trained team, the Group is proud of its well skilled members and expertise.

Manufacturing Activities of BSPC

- Steel Investment Castings
- Sand casting
- Gray Cast Iron Castings & Ductile Castings
- Precision CNC machining

Process

Why Investment Castings ?

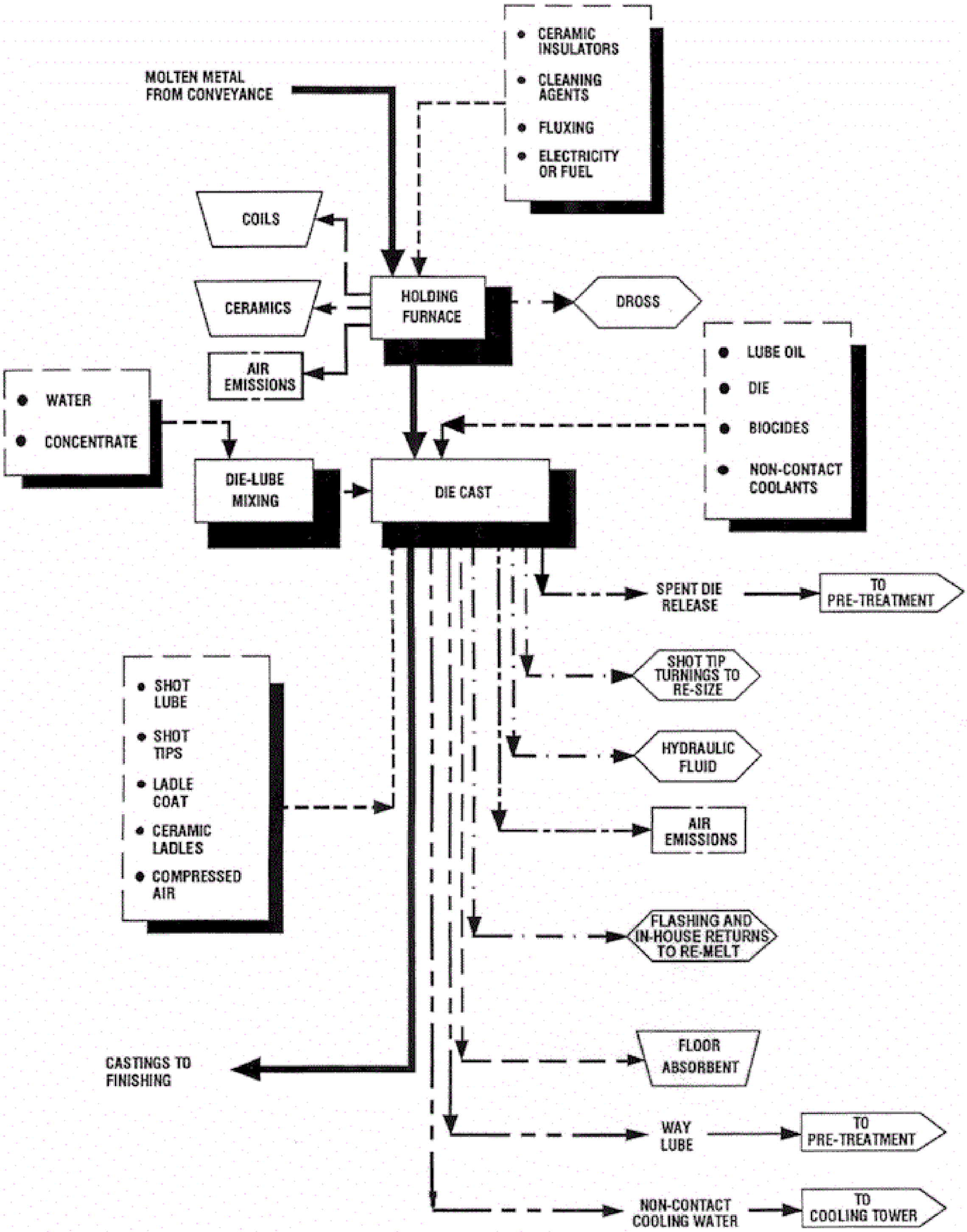
This is the only process in which you can see, touch & realize about your product before it cast.

By investment casting process, you can produce complicated shapes or near net shape that would be difficult or impossible to make by another manufacturing process by eliminating or substantially reducing the need for expensive machining or secondary operations.

History of Investment Castings

The history of lost-wax casting dates back thousands of years. Its earliest use was for idols, ornaments and jewellery, using natural beeswax for patterns, clay for the moulds and manually operated bellows for stoking furnaces. Examples have been found across the world in India's Harappan Civilisation (2500–2000 BC) idols, Egypt's tombs of Tutankhamun (1333–1324 BC), Mesopotamia, Aztec and Mayan Mexico, and the Benin civilization in Africa where the process produced detailed artwork of copper, bronze and gold.

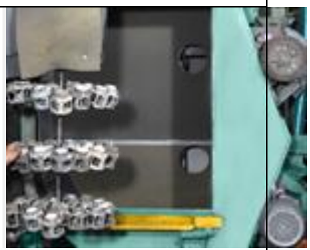
The earliest known text that describes the investment casting process (Schedula Diversarum Artium) was written around 1100 A.D. by Theophilus Presbyter, a monk who described various manufacturing processes, including the recipe for parchment. This book was used by sculptor and goldsmith Benvenuto Cellini (1500–1571), who detailed in his autobiography the investment casting process he used for the Perseus with the Head of Medusa sculpture that stands in the Loggia dei Lanzi in Florence, Italy.



Investment casting came into use as a modern industrial process in the late 19th century, when dentists began using it to make crowns and inlays, as described by Dr. D. Philbrook of Council Bluffs, Iowa in 1897. Its use was accelerated by Dr. William H. Taggart of Chicago, whose 1907 paper described his development of a technique. He also formulated a wax pattern compound of excellent properties, developed an investment material, and invented an air-pressure casting machine. In the 1940s, World War II increased the demand for precision net shape manufacturing and specialized alloys that could not be shaped by traditional methods, or that required too much machining. Industry turned to investment casting. After the war, its use spread too many commercial and industrial applications that used complex metal parts. Investment casting is an industrial process also called lost-wax casting, one of the oldest known metal-forming techniques from 5,000 years ago, when beeswax formed the pattern, to today's high-technology waxes, refractory materials and specialist alloys, the castings allow the production of components with accuracy, repeatability, versatility and integrity in a variety of metals and high-performance alloys. Lost foam casting is a modern form of investment casting that eliminates certain steps in the process.

The process is generally used for small castings, but has been used to produce complete aircraft door frames, steel castings of up to 300 kg (660 lbs) and aluminium castings of up to 30 kg (66 lbs). It is generally more expensive per unit than die casting or sand casting, but has lower equipment costs and more possibility to make your product very cost effect

Facilities:

			
Wax Injection Presses	Wax Pattern Assembly	Ceramic Coating	De-waxing with Auto Clave
			
Induction Furnace	Pneumatic Vibrator	Cutting & Heat Number Punching	Shot Blasting
			
Fettling	Coining Press	Heat Treatment Furnace	Pressure Testing

Quality & Assurance

The top management at Sumangal castings understands and constantly reiterates the fact that **Quality Comes First** in all its levels of operations. **Quality is the plinth of its operations** and the correct measures to maintain quality have helped the company maximize their efficiency, advance its growth and expand across borders consistently providing the world with highest quality products. The quality of its products is maintained with the use of high tech machineries and equipments like

			
Spectrometer	Certified Reference material for various grade	Pyrometers	Universal Testing Machine
			
Brinell Hardness Tester	Rockwell Hardness Tester	Impact Testing Machine	Various Height Gauges
			
Various Vernier Calipers	Radiography Testing	Dye Penetration Test	Radiation Meter

Products:

- Pumps
- Valves
- Cranes
- Automobile
- Machine Tools
- General Engineering
- Mechanical Seals
- Switch Gears
- Instrumentation
- Earth Moving

India's casting industry 19.2 billion by 2016

The industry is making a contribution of Rs 7,000 per ton produced to the national exchequer by way of excise and other levies.

Indian scenario

India has 4,600 foundry units, 80 per cent of them in the SME sector. Annual production is 7.4 million tons, approximately valued at $8 billion. The sector employs 0.5 million people directly and an additional 1.5 million people indirectly.

The industry is making a contribution of Rs 7,000 per ton produced to the national exchequer by way of excise and other levies. The sector exports touched $215 million in 2010/11. The manufacturing units are located in 12 identified and recognised clusters and in 20 other clusters.

Projected market size

While there is a growth expected in the auto-sector and the casting sector alike, the domestic market is set to surge at least three times by 2016. The auto sector alone should go up to $10 billion by 2016 and the casting industry is projected to be a $19.2 billion industry by 2016.

The global growth rate for castings follows approximately twice the global growth rate in GDP in absolute terms. So apart from the domestic demand, India's exports are to go up to $3 billion with a 20 per cent growth in direct exports.

Projected demand-supply gap

While there is a clear picture of demand side, the supply side is not prepared to make pro rata investments to take up the opportunity arising out of growth.

Even assuming that the past growth rate would continue, there will be a large demand-supply gap in excess of 11 million TPA by year 2016.

Diagnostic study of foundry sector

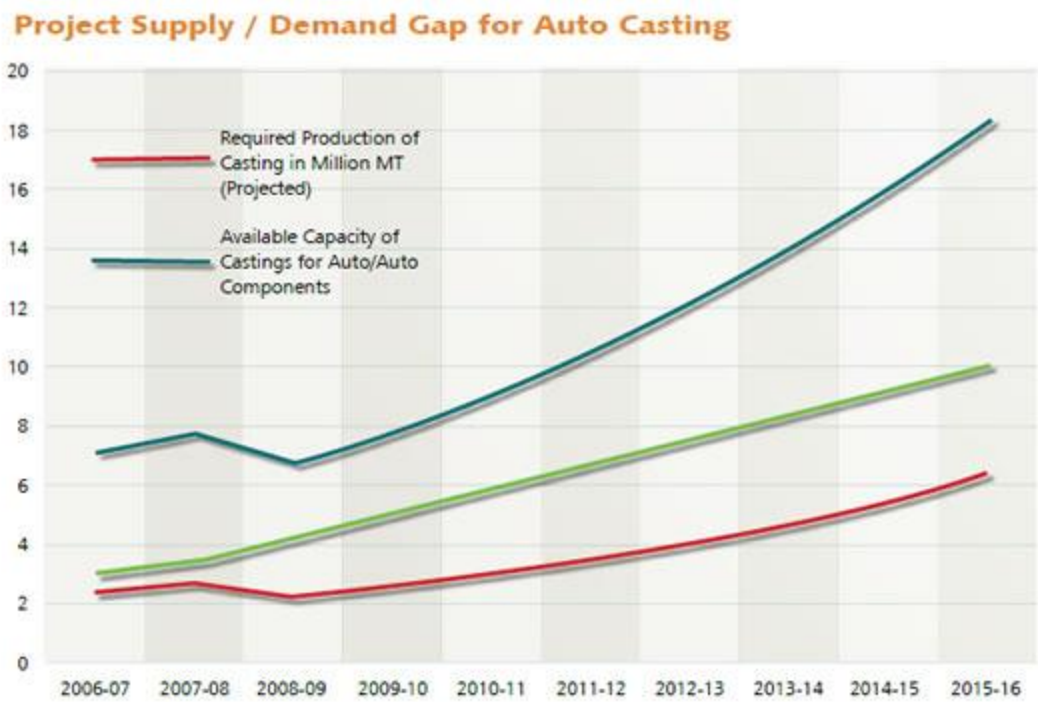
The data collected (from the 325 units and 100 personal interviews) are analysed with a diagnostic tree and the key findings based on them are summarised in the table below.

Determination of level of foundry

The industry has five distinct levels of units based on the size and sophistication each having its own dynamics of growth and plans. The market has also has four tiers and each level of unit has found its own market segment.

The foundries and markets are nicely dovetailed to each other in terms of price. But there is a disparity in terms of quality and quantity – between the market expectation and the supplies.

Project Supply / Demand Gap for Auto Casting



	Current Contribution	Current Value	Projected Growth by 2016
Auto Sector Demand	32% of demand	$2.5 billion (~2.18 MT)	$10 billion
Casting Industry for manufacturing	Over 90% of the local demand is locally met	~$18 billion	$19.2 billion (17-18 MT)
India Exports	15% of production	~$1 billion	$3 billion

The Indian foundry industry has lined up investments worth Rs 600 crore over the next few years as it expands and adapts environment-friendly measures to garner global marketshare.

A hefty chunk of the investment has been lined up include those in foundries located in West Bengal, traditional home of the domestic foundry industry and across the eastern region.

For starters, the Indian Institute of Foundrymen (IIF) a pan-India body representing the foundry sector,entered into a MoU with Japan Foundry Society, the largest organization of foundrymen in the world, for Green Business Foundry at the 61 stIndian Foundry Congress which got underway in the city on Sunday. The move will help implement cost-effective and eco-sensitive 'green' manufacturing standards across the country's $12-billion foundry industry.

"This comes at a time when a whopping Rs 600 crore worth of investment is expected to be invested in the foundry industry in West Bengal and the rest of eastern India," said Ravi Sehgal, chairman, organising committee of 61st IFC and IFEX 2013.

The three-day meet, being organized by IIF will have two concurrent events - IFEX 2013 and Cast India Expo. It is being attended by over 800 delegates from China, Germany, Italy and Japan, which are major hubs of the global foundry industry.

"The 61st IFC has a range of programmes and informative technical seminars addressed by reputed speakers, and an internationally attended exhibition of the latest machines and technology. Also, cast buyers of repute will be visiting to source their annual requirements, and industry representatives will be discussing their perceptions of the current scenario of the industry," Harsh K Jha, president, IIF said.

With 9.9 million tonne cast components, India is ranked among the top three globally with annual exports of around $2 billion. However, its share in the global market is below 2%.

"The theme for this year's meet is 'Building Brand India - The Casting Edge'.Foundrymen must thus make effort to offer the right product mix, quality and price in order to reach out to the world. Eastern India has taken up the challenge to set up new foundries, and with this the industry once again growing in strength in India.

Location of Major Clusters

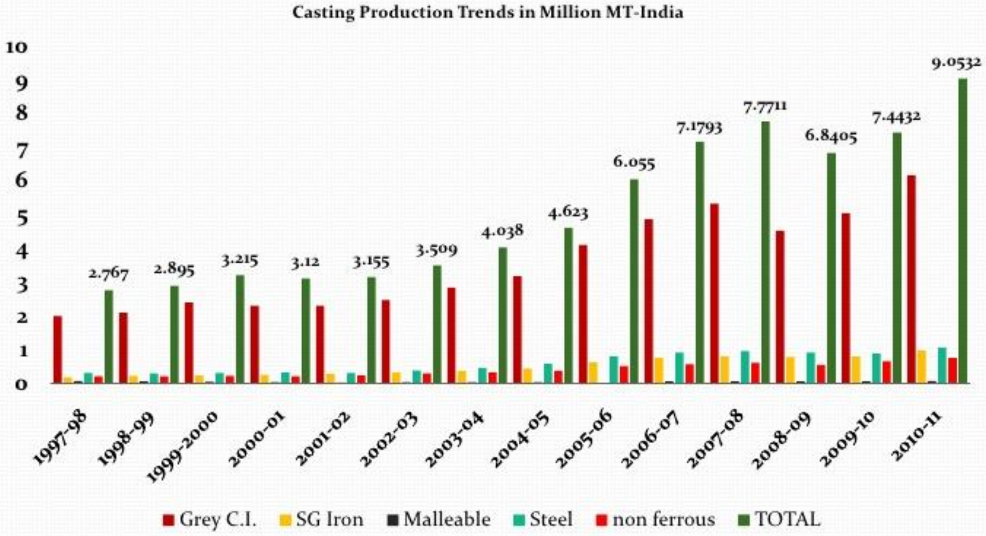
There are approx 4600 units in the country located in the areas marked in the map engaged in various types of metal castings

Indian Foundry Industry At a Glance

- Approx Units : 4600
- Production: 9 Million MT PA
- Employment: 500,000 Directly 150,0000 indirect
- Major Clusters: 19
- Productivity Per unit : 1950 MT/PA
- Avg Productivity/Man/PA: 20
- Max Productivity /Man/PA: 90

The Institute of Indian Foundrymen..

Castings Production Trends in Million MT-India

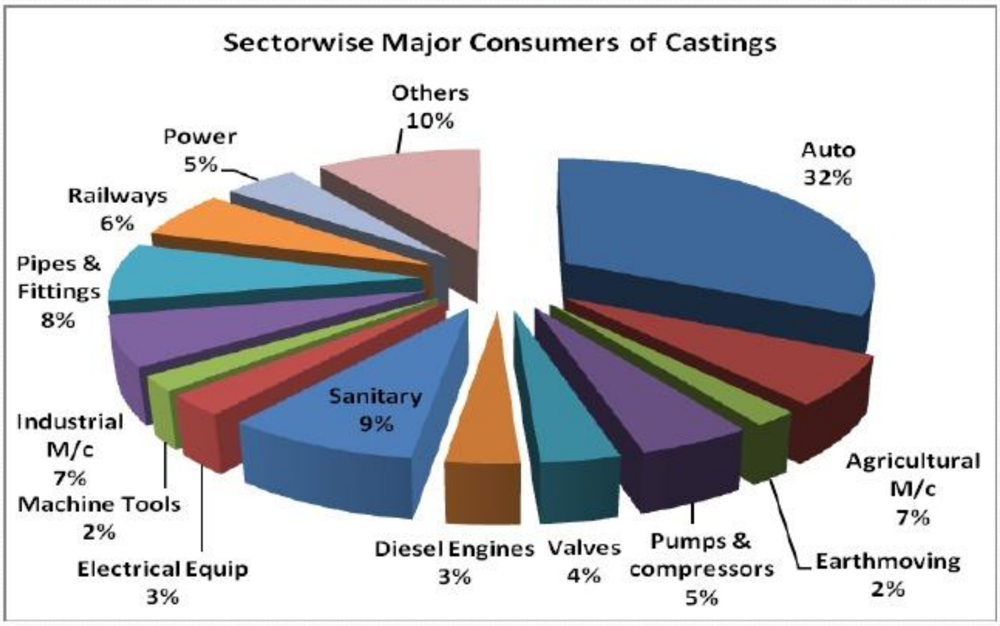


The Institute of Indian Foundrymen..

Exports

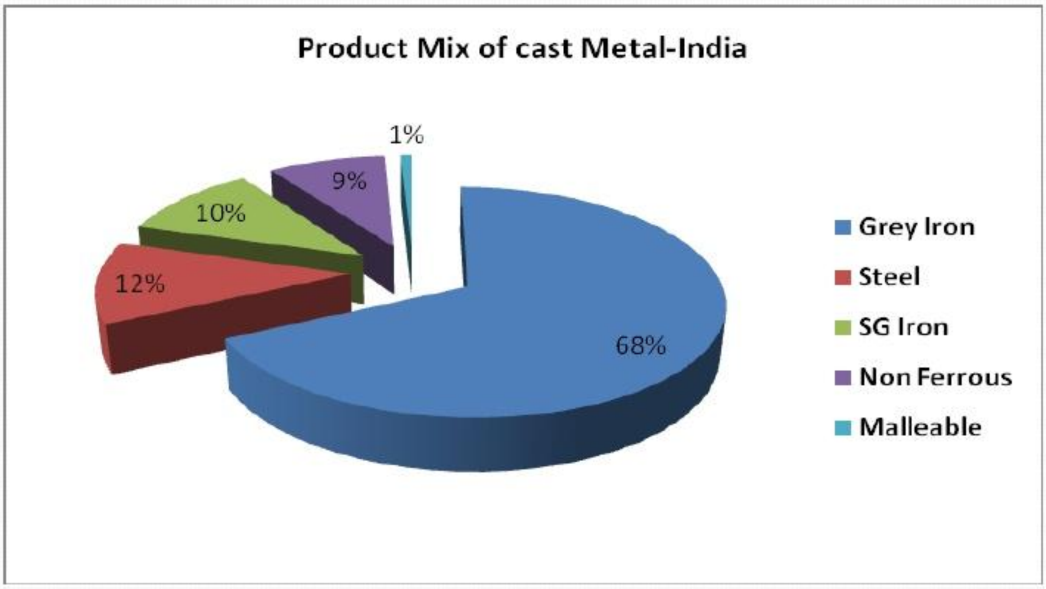
- Total engg exports during 2011 was USD 59.78 Billions ie an increase of 83.65% over previous year Castings export was approx Rs 6500 Cr in 2010-11 Vs Rs 4200 in 2009-2010
- The trend may not be sustainable next year as most Asian economies have increased interest rates to curb inflation & Coupled with rising commodity prices & crude will dampen demand
- However, most respondents expressed stagnant export market

Sectorwise Consumption of Castings

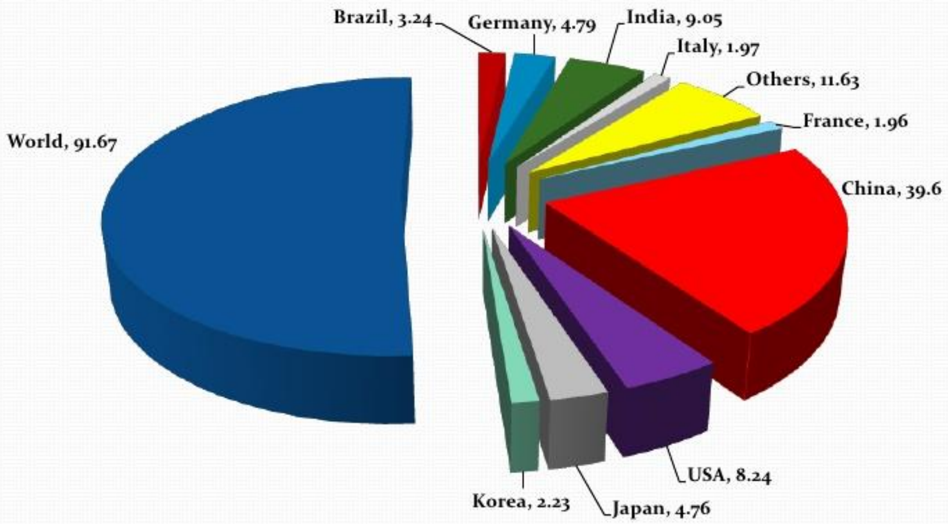


The Institute of Indian Foundrymen..

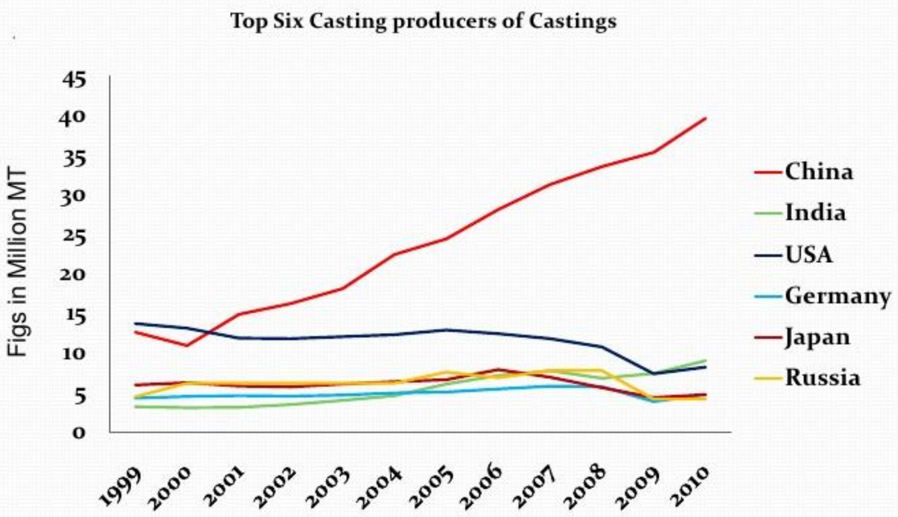
Product Mix



Share of Major Casting Producing Countries



Top 1-6 Casting Producers



Growth of Metal castings by –Major Countries by Production

While World production in 2010 grew by 14% as compared to previous years, Brazil emerged as fastest growing country as per census by Modern Castings

- Brazil 40.87%
- Germany 22.82%
- **India** **22.3%**
- Italy 17.96%
- China 12.18%
- USA 11.35%
- Japan 8.18%

Top 15 Produce >13%

Name of Unit	Production of Castings T/Year
	Year : 2010-11
ELECTROSTEEL CASTINGS LTD	275284
LANCO INDUSTRIES LTD.	123422
RAIL WHEEL FACTORY	106525
HINDUJA FOUNDRIES	88545
NELCAST LTD.	83469
TATA MOTORS JAMSHEDPUR + Pune	81274
BRAKES INDIA LTD.	75886
DCM ENGINEERING PRODUCTS	62327
SAKTHI AUTO COMPONENT LTD.	50000
COOPER FOUNDRY, SATARA	49200
MAHINDRA- HINODAY + Mahindra Auto	48719
JAYASWALS NECO LTD. +NECO CASTINGS	48085
ASHOK IRON WORKS	45112
KIRLOSKAR FERROUS INDUSTRIES LTD.	41905
WELCAST STEELS LTD.	38241

Foundry Industry -Foundries

- 50% respondants reported + 5-10% % growth in production in Q2 Vs Q1 2011-12 .While other 50% reported growth of + 10-15% in same period
- Exporters reported growth of + 5-10% in Q2 Vs Q1 2011-12
- Most reported order books + 5-10% in Q2
- Power availability improved in 2011-12 Vs 2010-11 though still unsatisfactory .The gap between demand & supply reduced in TN,AP,Punjab Rajasthan & Haryana
- Power generation grew by 9.3% in Apr-Sep 2011-12 Vs 4.1% in 2010-11

Equipment Manufacturers

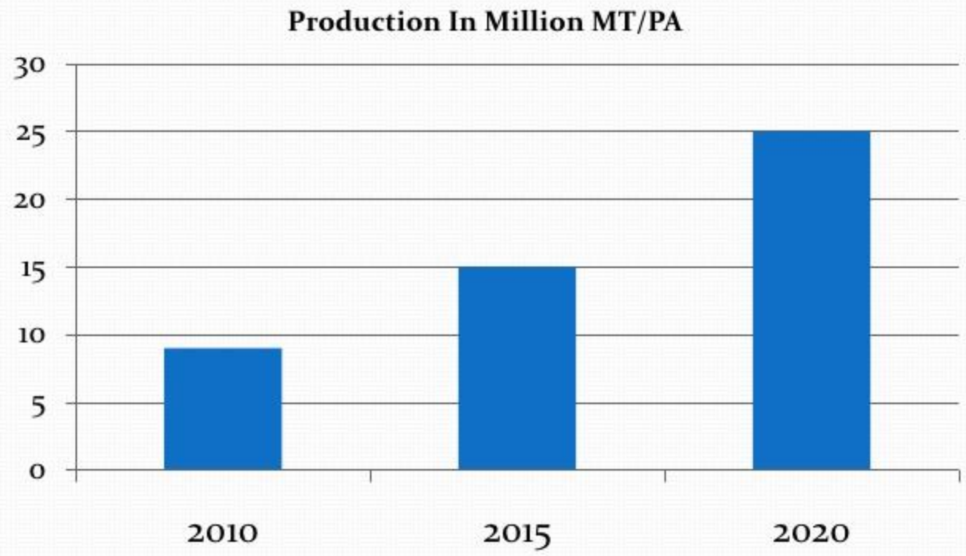
- Most respondants reported growth of +10-15% in Q2 11-12 Vs Q1
- Order books were Up by 5-10% reported by 50% respondants & up by 10-15% by balance in Q2 Vs Q1
- Reported raw material & power availability as satisfactory
- 75% Respondants reported export markets still growing while balance reported as stagnant

Foundry Material Manufacturers

- Most respondants have reported - ve growth of 5-10% in Q2 Vs Q1 while a few reported + ve growth during above period of 10-15%.
- Domestic markets for most is still growing @ 5-10% & satisfactory while exports are going down.

The Institute of Indian Foundrymen..

Vision For Indian Foundry Industry



Vision For Indian Foundry Industry